

SLIDE 1

ReturnGuard

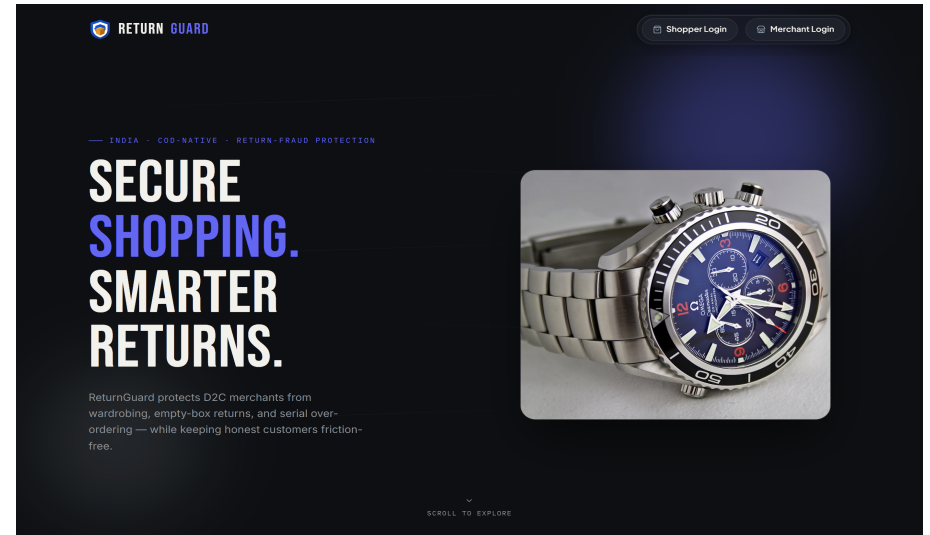
An Intelligent, COD-Native Risk Defense System for Indian D2C Commerce

Project Team: Team ReturnGuard | **Mentor Name: Thavayee**

PROJECT OVERVIEW

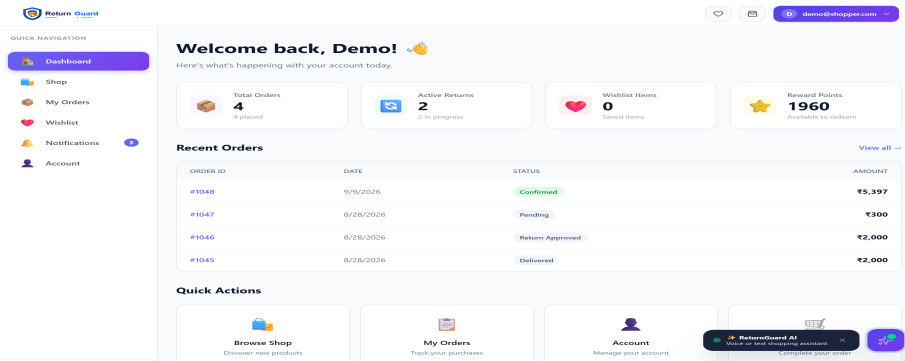
CORE SYSTEM CAPABILITIES & VALUE

- **Problem Context in Indian D2C:** Cash-on-Delivery (COD) accounts for 60%+ of orders, causing severe exposure where merchants pay 3x forward freight on abusive returns.
- **Dual-Portal Ecosystem:** Unified application integrating a customer shopping storefront and a merchant operations triage center.
- **Real-Time Scoring Engine:** Evaluates every order and return from **0 to 100 points** using passive device telemetry, history, and physical product checks.
- **Target Beneficiaries:** Built for small-to-mid Indian D2C brands (**\$50K–\$5M ARR**) in apparel, lifestyle, and electronics.
- **Frictionless by Default:** **85%+ of genuine orders & returns are auto-approved instantly** with zero customer friction; step-up checks trigger only on anomalies.
- **Strategic Value:** Converts reverse logistics from an uncontrolled margin drain into a secure, revenue-protecting operational asset.



Live Application Screen: Storefront Landing Page & Shopper Commerce Experience

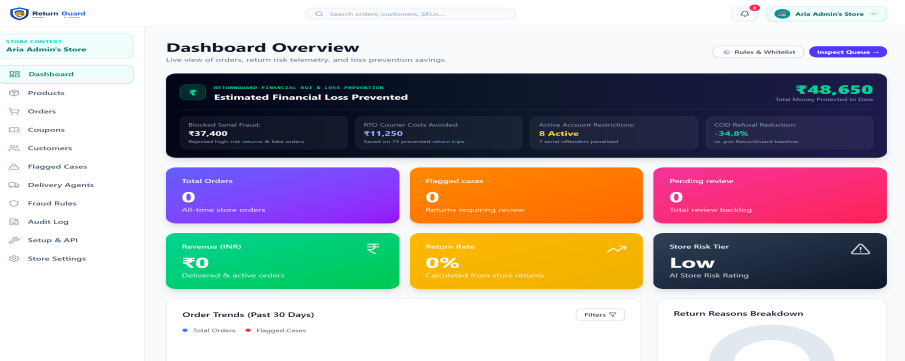
PLATFORM ARCHITECTURE: SHOPPER & MERCHANT DASHBOARDS



Shopper Portal: Order History, Reward Points & 1-Click Return/Exchange Hub

SHOPPER STOREFRONT & SELF-SERVICE PORTAL

- **1-Click Return Initiation:** Frictionless self-service return and exchange requests directly from customer order history.
- **Automated Smart Exchanges:** Recommends instant size/color swaps over cash refunds, reducing reverse freight while preserving the sale.
- **Real-Time Timeline Tracking:** Live status updates on pickup scheduling, doorstep courier inspection, and refund processing.
- **Instant Friction-Free Approvals:** 85%+ of genuine returns auto-cleared immediately with zero administrative delays.



Merchant Portal: Executive Risk KPIs, Flagged Returns Queue & Fraud Triage

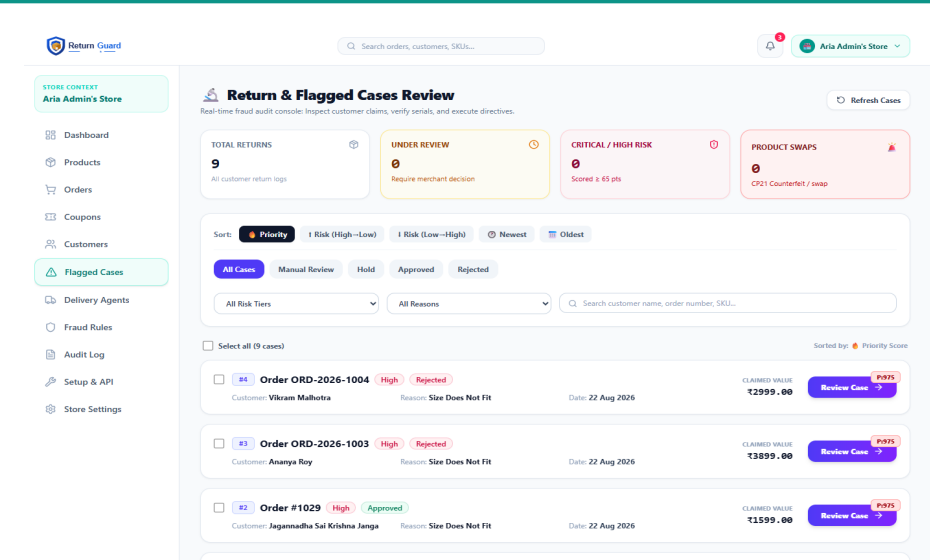
MERCHANT OPERATIONS COMMAND CENTER

- **Executive KPI Surveillance:** Real-time visibility into overall return rates, prevented fraud loss values, and risk tier distributions.
- **Algorithmic Anomaly Triage:** Automated 0–100 scoring funneling suspicious claims into a prioritized review queue.
- **Configurable Fraud Policy:** Dynamic sliders to adjust sensitivity per product category, festive wardrobing rules, and serial tracking.
- **Custody Chain & Logistics:** Doorstep digital witnessing, courier photo verification, and delivery agent collusion analytics.

CHALLENGES & SOLUTIONS

FRAUD CHALLENGES & ENGINEERING SOLUTIONS

- **Challenge 1 — Wardrobing & Festive Rental Abuse:** Shoppers purchase high-value ethnic apparel for weddings/Diwali, wear once, and return for full refunds.
→ **Solution (CP4):** Automated festive calendar risk tightening, mandatory tag-intact verification, and pre-return condition checks.
- **Challenge 2 — Size Bracketing Multi-Orders:** Buyers order 3–4 sizes of the same SKU to try at home and return the rest, inflating reverse freight by 300%.
→ **Solution (CP10):** Cart-level multi-size detection with instant size exchange incentives over cash returns.
- **Challenge 3 — Electronics Hardware Swapping:** High-value electronics returned with missing accessories, clone devices, or dummy ballast weights.
→ **Solution (CP17):** Mandatory serial/IMEI verification between dispatch and return with +50 pt critical risk flagging.
- **Challenge 4 — Courier & Driver Collusion:** Delivery drivers falsely marking items as 'customer rejected' or siphoning returned goods en route.
→ **Solution (CP24):** Geo-stamped doorstep signatures, tamper-evident seals, and driver route anomaly analytics.

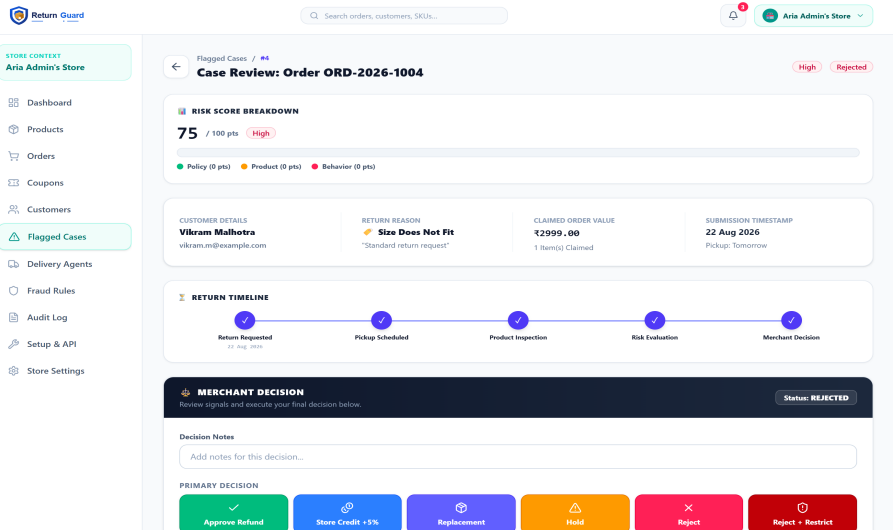


Live Application Screen: Flagged Cases Queue (Actionable Risk Scoring & Resolution Workflow)

KEY FEATURES

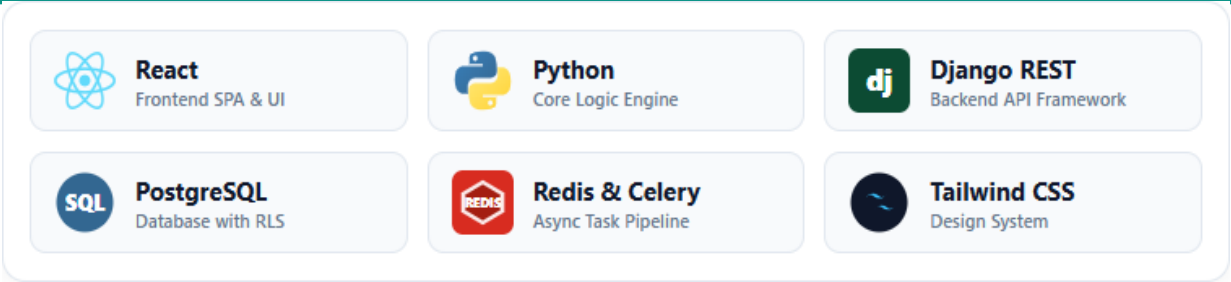
KEY IMPLEMENTED ARCHITECTURAL FEATURES

- **Customer & Risk Profile Context:** Real-time shopper telemetry displaying customer identity, order history, active return rate, and flagged risk tier (High Risk • 78 pts).
- **5-Step Return & Risk Engine:** Automated lifecycle progression: **Return Requested** → **Pickup Scheduled** → **Product Inspection** → **Risk Evaluation** → **Merchant Decision**.
- **1-Click Merchant Decision Panel:** Direct operational actions: **Approve Refund**, **Store Credit** (+5%), **Replacement**, **Hold**, **Reject**, or **Reject + Restrict**.
- **4-Tier / 28 Risk Checkpoints:** Comprehensive evaluation pipeline covering Pre-Scoring Gates, Physical Attributes, Behavioral Biometrics, and Composite Decision Rules.
- **6-Level Escalation Ladder:** Proportionate customer friction scaling from instant auto-approval to OTP challenge, COD lock, and account termination.
- **Full Audit & Case Documentation:** Structured decision notes capture and immutable audit logging for fraud disputes and chargeback defense.



Live Application Screen: Flagged Case Review (Customer Profile, 5-Step Return Engine & Decision Panel)

TECHNOLOGY STACK



FRONTEND & CLIENT ARCHITECTURE

- **React & Vite SPA:** Modular component architecture powering both shopper storefront and merchant command center with hot module replacement.
- **Tailwind CSS Design System:** Modern utility styling delivering responsive layouts, risk status badges, and accessible tables.
- **Lucide React Icons & Context API:** Clean SVG iconography with lightweight global state management for auth, cart, and alerts.
- **Resilient API Client:** Axios HTTP layer featuring automatic JWT token refreshing and error boundary fallback states.

BACKEND, DATABASE & TASK PIPELINE

- **Python & Django REST Framework:** Robust RESTful API architecture running the 28-checkpoint scoring algorithm with sub-second execution.
- **PostgreSQL with Row-Level Security:** Relational database with strict tenant isolation, foreign keys, and ACID compliance.
- **Redis & Celery Worker Queue:** Distributed asynchronous background pipeline handling risk scoring and automated alerts.
- **Security & Invoicing Engine:** HMAC SHA-256 webhook signatures, encrypted telemetry, and ReportLab PDF invoice generation.

CONCLUSION

CORE SYSTEM DELIVERABLES

- **Complete Working Platform:** Dual-portal e-commerce application integrating a customer storefront with a real-time merchant operations command portal.
- **28 Algorithmic Checkpoints:** Comprehensive multi-tier scoring pipeline analyzing behavioral, cultural, physical, and logistics checks.
- **6-Level Escalation Ladder:** Dynamic, proportional friction scaling from automated approval to doorstep OTP, COD restriction, and blacklisting.
- **Production API Architecture:** Scalable Django REST backend with asynchronous task workers and PostgreSQL Row-Level Security isolation.
- **Full Audit & Case History:** Immutable tracking logs capturing every risk signal, customer action, and merchant resolution.

BUSINESS VALUE & QUANTIFIED IMPACT

- **85%+ Instant Approvals:** Genuine customers experience zero friction with instant auto-approval on legitimate orders and return claims.
- **70% Cut in Wardrobing Abuse:** Cultural and seasonal calendar rules eliminate organized festive garment rental fraud during weddings and festivals.
- **Reverse Freight Recovery:** Eliminates unnecessary 3x forward logistics fees on abusive multi-size orders and merchandise swapping.
- **Protects Merchant Margin:** Empowers small-to-mid Indian D2C retailers to offer safe COD payments without risking inventory bankruptcy.
- **Granular Policy Customization:** Store operators can adjust risk sensitivity sliders dynamically per product category to optimize margins.

FUTURE SCOPE

1. MOBILE EDGE COMPUTER VISION

- **Real-Time Doorstep Scanning:** Deploy lightweight on-device AI vision models to couriers' handheld devices for real-time fabric texture analysis, tag integrity verification, and wear detection prior to acceptance.

2. CONSORTIUM FRAUD GRAPH NETWORK

- **Cross-Merchant Identity Mesh:** Privacy-preserving zero-knowledge graph network that pools anonymized risk telemetry across independent Indian D2C stores to preemptively flag serial fraudsters.

3. MANIFEST V3 CHROME OPS EXTENSION

- **Embedded E-Commerce Integration:** Browser toolbar extension injecting ReturnGuard's 0-100 risk badges and 1-click triage buttons directly inside Shopify, WooCommerce, and Magento order management panels.

4. AUTOMATED UPI PENNY-DROP & REFUNDS

- **Frictionless Micro-Settlements:** Algorithmic bank account verification via  NPCI UPI penny-drops, enabling automated fractional payouts and dynamic deductions for missing product accessories or tags.

SLIDE 9

THANK YOU

ReturnGuard: AI-Assisted Return-Fraud Detection Platform

Open for Questions, Architectural Inquiries & Technical Discussion